Improving thin-vessel segmentation in fundus images: dual attention, a topology-aware loss and higher resolution for CAR-UNet on the multi-disease FIVES dataset

Abhishek Karthik J, Abhishek Ramesh, Krish S and Theerth Krish S

Amrita School of Artificial Intelligence, Amrita Vishwa Vidyapeetham, Coimbatore, Tamil Nadu, India

ARTICLE INFO

Keywords:
Retinal vessel segmentation
Fundus images
CAR-UNet
Attention mechanisms
clDice
FIVES dataset

ABSTRACT

Retinal vessel maps support screening for diabetic retinopathy, glaucoma and other eye diseases, but the thinnest capillaries, where early changes appear, remain the main source of segmentation errors. The Channel Attention Residual U-Net (CAR-UNet) reports high accuracy on small single-source benchmarks, yet it has not been tested on large multi-disease data, and its channel-only attention and pixel-wise loss do little to recover thin or broken vessels. We replicated CAR-UNet on the public FIVES dataset of 800 fundus images (normal eyes, age-related macular degeneration, diabetic retinopathy and glaucoma) with a disease-stratified split, three training seeds, all 200 test images and paired Wilcoxon tests, and then extended it with four changes aimed at thin vessels: spatial attention alongside channel attention, a topology-aware centreline Dice (clDice) loss, a working resolution of 1024×1024 instead of 512×512 pixels, and a decision threshold selected on validation data. The replicated model reached a Dice score of 0.8382 ± 0.0031 and outperformed a plain U-Net on 154 of 200 test images. The extended model reached a Dice score of 0.8436 ± 0.0027, an intersection over union of 0.7379 and a clDice of 0.8751, improved vessel connectivity on 175 of 200 test images ($p < 10^{-24}$), and gained 2.2 Dice points against the 1024×1024 ground truth. Targeting thin vessels therefore improves CAR-UNet consistently on multi-disease data, although glaucoma and very low-quality images remain difficult.

Group members (22AIE437 Medical Image Processing, final project)

Abhishek Karthik J	CB.SC.U4AIE23010	Abhishek Ramesh	CB.SC.U4AIE23011
Krish S	CB.SC.U4AIE23040	Theerth Krish S	CB.SC.U4AIE23071

1. Introduction

The retina offers a rare, non-invasive view of the body's microvasculature. A single fundus photograph shows arteries, veins and capillaries whose width, tortuosity, branching pattern and continuity change with diabetic retinopathy (DR), glaucoma, age-related macular degeneration (AMD) and hypertension. Measuring these changes requires a vessel map, a pixel-wise segmentation of the vascular tree. Producing such maps by hand is slow and tedious, and different experts disagree, especially on the faint capillaries at the ends of the tree. Early pathological changes appear precisely there, which makes automatic vessel segmentation an attractive target for medical image analysis.

The task is harder than it first appears. In FIVES, vessels make up only about 7% of the image (about 10% of the retinal field of view), so any method is pushed towards predicting background. Illumination varies strongly across the image, the optic disc and lesions such as exudates and haemorrhages produce high-contrast structures that resemble vessels, and the thinnest capillaries are only one or two pixels wide once the image is resized to a manageable resolution. Classical approaches based on vesselness filters (Frangi et al., 1998) handle the large vessels but break down on these thin, low-contrast structures. Deep convolutional networks have largely replaced them (Liskowski and Krawiec, 2016), with the U-Net encoder–decoder architecture (Ronneberger et al., 2015) becoming the standard backbone. Many variants add attention, among them Attention U-Net (Oktay et al., 2018), SA-UNet (Guo et al., 2021b) and CAR-UNet (Guo et al., 2021a), the base paper of this project. CAR-UNet replaces the convolution blocks of U-Net with residual blocks that contain an efficient channel-attention module, and applies the same attention to the skip connections.

CAR-UNet was evaluated on small, single-source benchmarks with 20 to 40 images, such as DRIVE (Staal et al., 2004), and only with accuracy, sensitivity, specificity and AUC. It is therefore unclear how it behaves on larger, multi-disease data, and whether its design really recovers the thin vessels that matter clinically. Its design also leaves two

gaps for thin vessels: its attention re-weights feature channels but not image locations, and its cross-entropy loss barely penalises a vessel that is broken into pieces. Our replication confirms that most of its remaining errors are thin capillaries that are missed or fragmented.

In this work we argue that the remaining errors of CAR-UNet on multi-disease data come mainly from thin vessels, and that changes aimed specifically at them (spatial attention, a topology-aware loss and a higher working resolution) improve the model consistently. On FIVES, these changes raise the Dice score from 0.8382 to 0.8436 and the centreline Dice from 0.8626 to 0.8751, with gains in all three training seeds. The main contributions are:

- We replicate CAR-UNet on FIVES (Jin et al., 2022), an 800-image dataset with four disease groups, under a strict and reproducible protocol, and compare it with U-Net.
- We identify five weaknesses of the base model and its evaluation, and propose four targeted modifications: dual (channel + spatial) attention, a topology-aware clDice loss, a higher working resolution, and a validation-selected decision threshold.
- We evaluate every model with three training seeds on all 200 test images, report Dice, IoU, clDice and the paper's metrics, test every difference with paired Wilcoxon tests, and quantify the contribution of each modification in an ablation study.
- We analyse the results per disease and on thin vessels, and report the cases where our approach does not help.

The rest of the report is organised as follows. Section 2 reviews related work, Section 3 describes the base paper and our replication, Section 4 presents the methodology and the proposed modifications, Section 5 reports the results, Section 6 discusses them, and Section 7 concludes.

2. Related work

Vessel segmentation. Early methods enhanced tubular structures with Hessian-based filters (Frangi et al., 1998) or matched filters and then thresholded the response. Supervised deep networks now dominate. Liskowski and Krawiec (2016) showed that convolutional networks trained on image patches outperform hand-crafted pipelines, and the U-Net (Ronneberger et al., 2015), with its contracting path, expanding path and skip connections, has become the most common backbone because it combines context with precise localisation. Most of these methods, however, were developed on a few dozen images, and their remaining errors concentrate on thin vessels.

Attention and residual learning. Residual connections (He et al., 2016) make deeper networks easier to optimise. Channel attention, introduced with squeeze-and-excitation networks (Hu et al., 2018) and made lightweight by ECA-Net (Wang et al., 2020), reweights feature channels. CBAM (Woo et al., 2018) combines channel attention with spatial attention that reweights image locations. In retinal imaging, Attention U-Net (Oktay et al., 2018) gates the skip connections, SA-UNet (Guo et al., 2021b) uses spatial attention, and CAR-UNet (Guo et al., 2021a) builds residual blocks around a modified efficient channel attention (MECA). CAR-UNet therefore decides which feature maps matter, but not where in the image the vessels lie.

Losses and topology. Cross-entropy and the soft Dice loss (Milletari et al., 2016) are the usual training objectives, but both count pixels and are almost blind to whether a thin vessel is connected. The clDice loss (Shit et al., 2021) compares the skeletons of prediction and ground truth and was designed for tubular structures such as vessels. CAR-UNet was trained with cross-entropy alone and does not use such a term.

Datasets. DRIVE (Staal et al., 2004) and similar benchmarks contain a few dozen images from a single source. FIVES (Jin et al., 2022) provides 800 images at 2048×2048 pixels, with expert-consensus annotations and a balanced mix of normal, AMD, DR and glaucoma cases, which makes it a more demanding testbed. To our knowledge, CAR-UNet has not been evaluated on it.

3. Base paper and replication

3.1. The CAR-UNet architecture

CAR-UNet (Guo et al., 2021a) keeps the U-shaped layout of U-Net (Fig. 1). Each encoder and decoder stage, and the bottleneck, is a *channel attention double residual block* (CADRB): two 3×3 convolutions with batch normalisation,

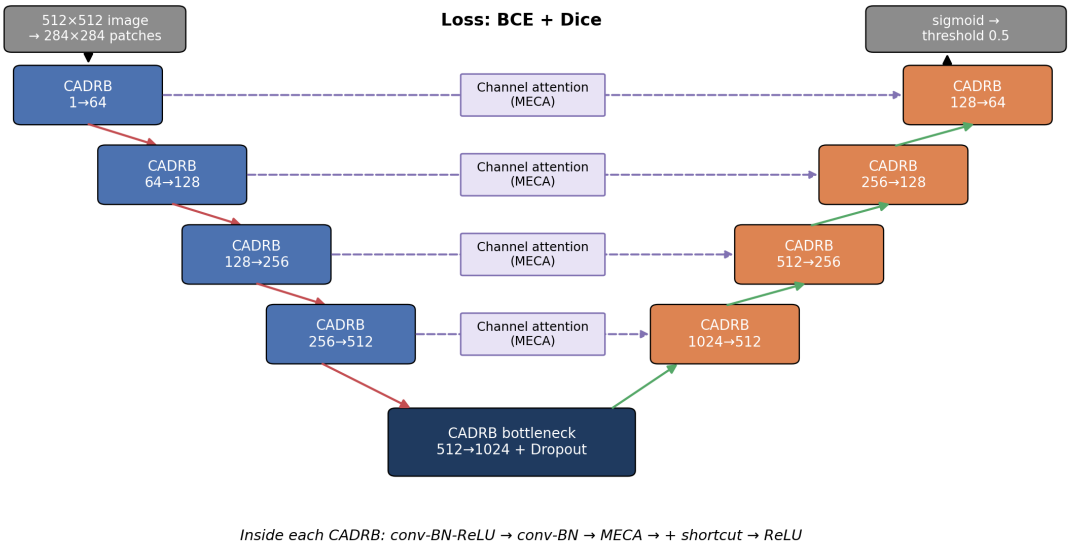


Figure 1: CAR-UNet as implemented in this work. A 284×284 input patch is mapped to a 100×100 vessel probability map with valid convolutions. Blue: encoder blocks; orange: decoder blocks; dashed arrows: skip connections with channel attention (MECA).

Table 1
Results reported in the base paper (Guo et al., 2021a).

Dataset	Accuracy	Sensitivity	Specificity	AUC
DRIVE	0.9699	0.8135	0.9849	0.9852
CHASE_DB1	0.9751	0.8439	0.9839	0.9898
STARE	0.9743	0.8445	0.9850	0.9911

followed by channel attention, plus a shortcut connection that adds the block input to its output. The channel attention (MECA) pools every feature channel to one number, passes the resulting vector through a small 1D convolution across neighbouring channels and a sigmoid, and multiplies each channel by its weight:

$$\mathbf{F}' = \sigma\big(\mathrm{C1D}_k\big(\mathrm{GAP}(\mathbf{F})\big)\big) \odot \mathbf{F}, \tag{1}$$

where GAP is global average pooling, $\mathrm{C1D}_k$ a 1D convolution with kernel size k and σ the sigmoid. The same attention is applied to the encoder features on every skip connection before they are concatenated with the decoder features. The paper trains with binary cross-entropy and Adam and reports the results in Table 1; it does not report Dice or IoU.

3.2. Our implementation and its differences from the paper

We implemented CAR-UNet in PyTorch. To stay consistent with our earlier U-Net baseline we kept the U-Net design with valid convolutions (284×284 input, 100×100 output), 64 base filters and dropout in the bottleneck. The original paper uses 16 base filters, DropBlock (Ghiasi et al., 2018) and a plain binary cross-entropy loss, and its MECA pools each channel by both average and maximum. We train with a combined cross-entropy and Dice loss (Section 4.4). Our replicated model uses average pooling only, as in Eq. (1), with an adaptive kernel size $k = |(\log_2 C + 1)/2|_{\mathrm{odd}}$ (minimum 3). The replicated model has 32.4 M parameters, against 31.0 M for our U-Net. We state these differences explicitly because they affect absolute numbers.

3.3. Replication protocol

A first version of our experiment produced results that turned out to be unreliable, for three main reasons. Only 50 of the 200 test images were scored, and because the filenames were sorted as text, 45 of these 50 were glaucoma

Table 2
Reported versus replicated results. Our results: 200 FIVES test images, mean ± standard deviation over three training seeds. n/r: not reported.

Model	Dataset	Dice	IoU	Accuracy	Sensitivity	Specificity	AUC
CAR-UNet (paper)	DRIVE	n/r	n/r	0.9699	0.8135	0.9849	0.9852
CAR-UNet (paper)	CHASE_DB1	n/r	n/r	0.9751	0.8439	0.9839	0.9898
CAR-UNet (ours)	FIVES	0.8382 ± 0.0031	0.7295 ± 0.0043	0.9728	0.8310	0.9855	0.9860
U-Net (ours)	FIVES	0.8344 ± 0.0034	0.7246 ± 0.0050	0.9723	0.8247	0.9852	0.9847

images. The validation set contained no DR or glaucoma images. The two models were trained with different settings. We fixed all of these before running any of the experiments reported here. Both models are trained by one script with identical settings (Section 4.4), the validation set is stratified by disease, every model is trained with three seeds, and all 200 test images are scored.

3.4. Replicated results

Table 2 compares the published results with our replication. A one-to-one comparison is not possible because the datasets differ: the paper used small single-source sets, whereas we use the 800-image multi-disease FIVES dataset. Still, our accuracy, specificity and AUC fall in the same range as the published values, and our sensitivity lies between the paper's DRIVE and CHASE_DB1 results. More importantly, the replication reproduces the paper's main claim: CAR-UNet is better than U-Net, with a higher Dice on 154 of the 200 test images (paired Wilcoxon test, $p = 2.3{\times}10^{-14}$). The margin is small (+0.38 Dice points) and varies between seeds (+1.06, −0.07 and +0.14 points). Fig. 2 shows typical predictions. Thick vessels are traced almost perfectly, while most errors are thin capillaries that are missed or fragmented.

4. Methodology

Fig. 3 summarises the full pipeline, from the raw fundus images to the evaluated vessel maps.

4.1. Dataset and split

FIVES (Jin et al., 2022) contains 800 colour fundus images of 2048×2048 pixels from 573 patients, annotated by a consensus of ophthalmologists and trained medical staff, with 200 images each of normal eyes, AMD, DR and glaucoma. It comes with an official split into 600 training and 200 test images (50 per group). We divide the 600 training images into 480 for training and 120 for validation, stratified by disease (30 validation images per group) with a fixed seed. The split is made per image, so that no patch of a validation image is ever seen during training. The test set is used only once, for the final evaluation; every choice (best epoch, threshold) is made on the validation set. Images are resized to 512×512 for the baseline models and to 1024×1024 for the high-resolution model, using area interpolation for images and nearest-neighbour interpolation for masks, so that the masks stay binary.

4.2. Preprocessing

Every image goes through the same classical pipeline (Fig. 4). We keep the green channel, where haemoglobin absorbs most strongly and vessels have the highest contrast. Contrast-limited adaptive histogram equalisation (CLAHE) (Zuiderveld, 1994) (clip limit 2.0, 8×8 tiles) compensates for uneven illumination, and an edge-preserving bilateral filter (Tomasi and Manduchi, 1998) ($d = 5$, $\sigma = 75$) removes noise without blurring the vessel edges. Intensities are then scaled to $[0, 1]$. A field-of-view (FOV) mask of the circular retina is computed by thresholding and morphological filtering; all metrics are computed inside it.

4.3. Patch sampling, augmentation and full-image inference

Following U-Net, we use valid convolutions: the network shrinks its input by 184 pixels, so that a 284×284 input patch yields a 100×100 prediction of its centre, and every output pixel sees 92 pixels of real context on each side. In every epoch we draw six new random patches per training image, centred inside the FOV (2,880 patches per epoch). Each patch is augmented with random horizontal and vertical flips, 90° rotations and, with probability 0.35, an elastic

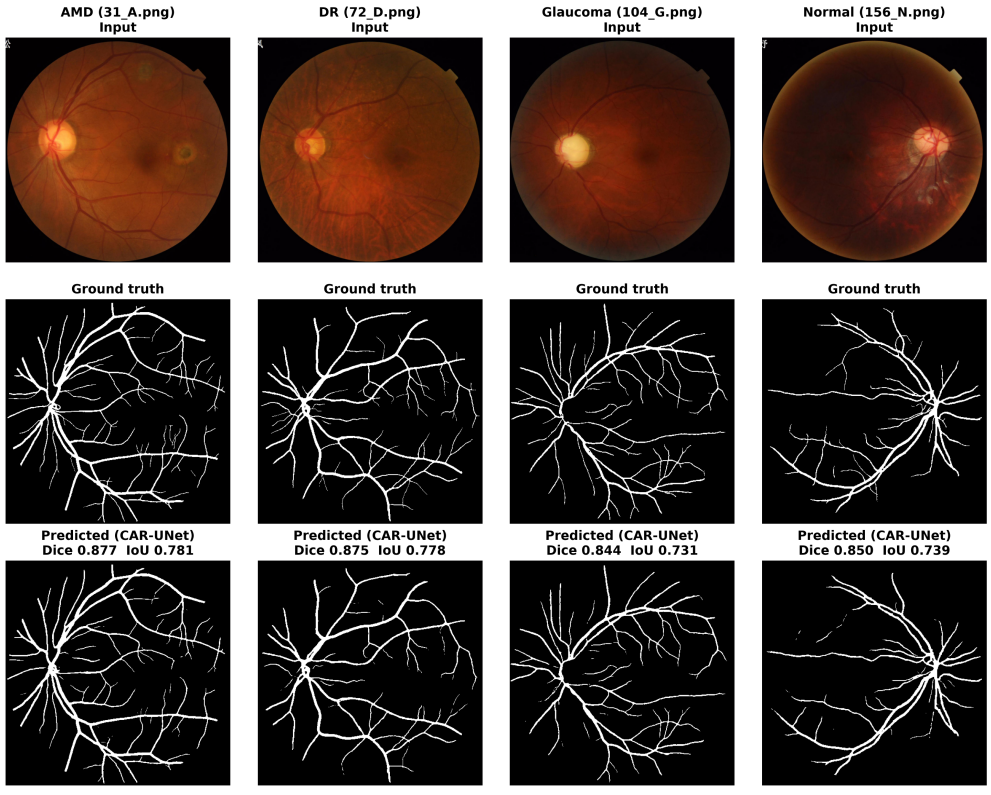


Figure 2: Predictions of the replicated CAR-UNet (seed 0) on one test image per disease group, chosen automatically as the image with the median Dice of its group. Rows: input fundus image, expert ground truth, predicted vessel mask with its Dice and IoU.

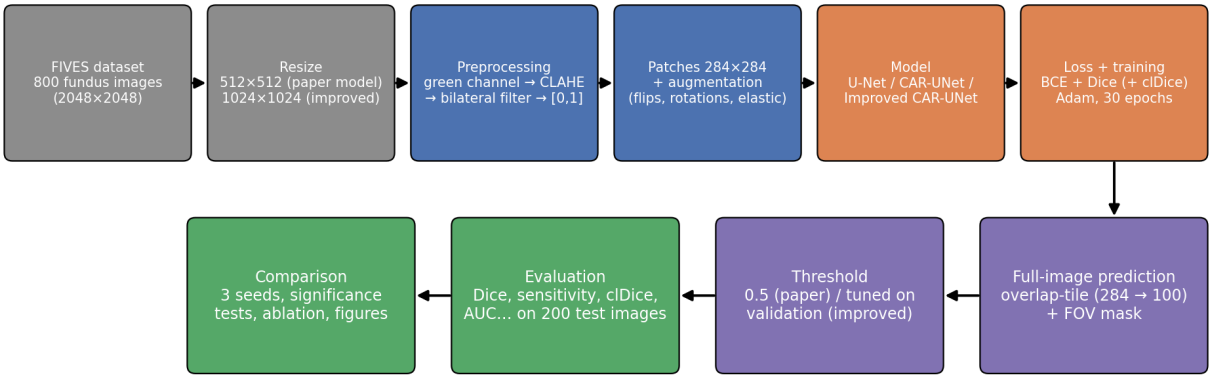


Figure 3: Overview of the pipeline used for all models.

deformation ($\alpha = 25$, $\sigma = 4$). All random generators are seeded. For full images we use the overlap-tile strategy: the image is mirror-padded by 92 pixels and covered by a grid of 100×100 output tiles, each predicted from its 284×284 input window. Pixels outside the FOV are set to zero.

4.4. Training

All models are trained by the same script with identical settings (Table 3). The loss of the baseline models is $\mathcal{L} = 0.5\,\mathcal{L}_{\mathrm{BCE}} + 0.5\,\mathcal{L}_{\mathrm{Dice}}$, which combines the stable gradients of binary cross-entropy with the imbalance-robust soft

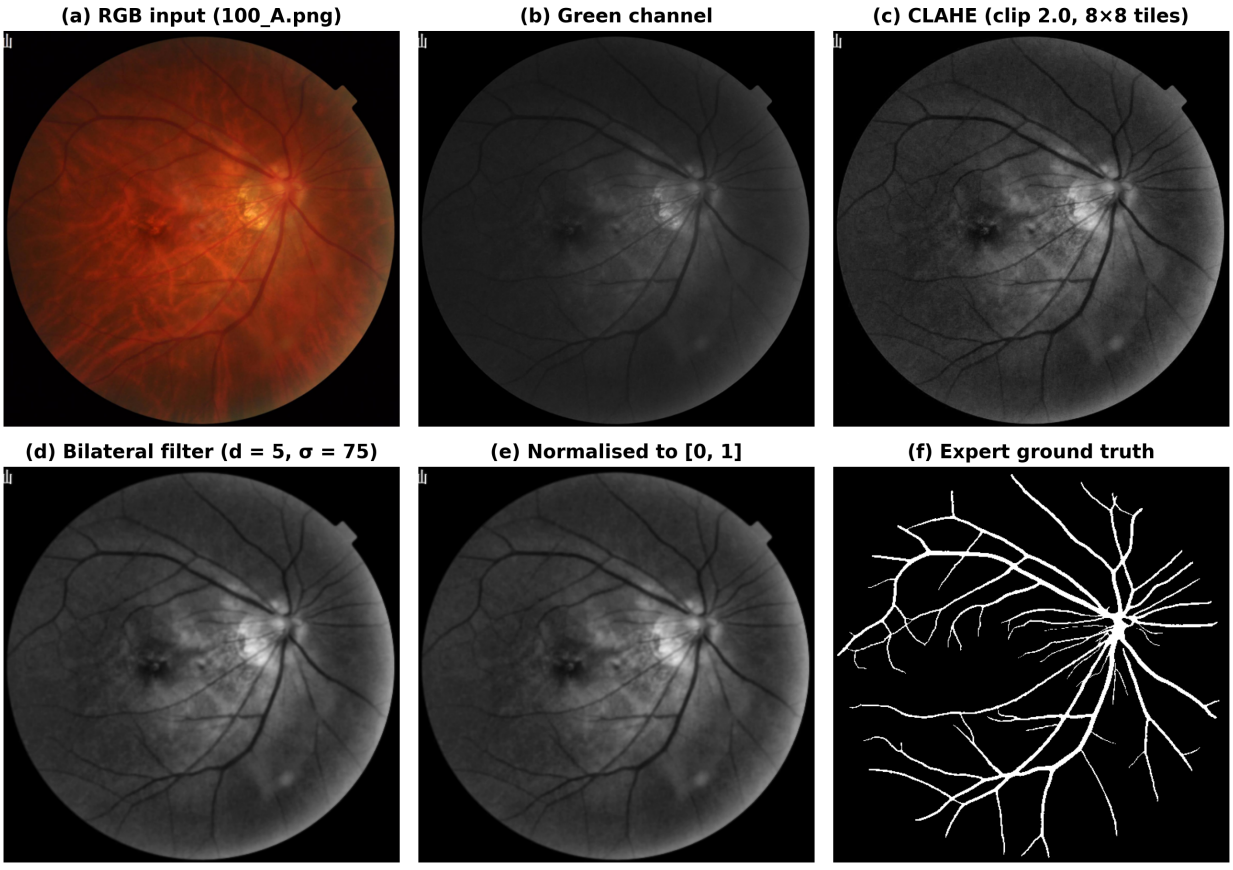


Figure 4: Preprocessing of a FIVES training image: RGB input, green channel, CLAHE, bilateral filter, normalised image and the expert ground truth.

Table 3
Training settings, identical for every model.

Framework	PyTorch 2.6, CUDA 12.4
Optimiser	Adam (Kingma and Ba, 2015), learning rate 10^{-3}, weight decay 10^{-5}
Schedule	cosine annealing over 30 epochs (Loshchilov and Hutter, 2017)
Batch	8 patches of 284×284 pixels; 6 new patches per image per epoch
Epochs	30, best validation-Dice checkpoint
Seeds	3 per model (0, 1, 2)
Hardware	NVIDIA RTX 4050 laptop GPU (6 GB)
Training time	≈71 min (U-Net), ≈91 min (CAR-UNet), ≈195 min (improved, 1024^2) per run

Dice loss (Milletari et al., 2016). After every epoch the model is evaluated on the 120 full validation images, and the epoch with the highest mean validation Dice is kept. Fig. 5 shows the training behaviour of the baseline models.

4.5. Implementation

All models are implemented in PyTorch, and one code base is used for every experiment (Table 4). Each run follows the same chain: images are loaded and preprocessed, random 284×284 patches are drawn, the model and loss are applied, the full validation or test images are predicted with the overlap-tile strategy, and the metrics are computed inside the FOV. Per-image results of every run are stored, so that all tables and significance tests in this report can be regenerated from the saved files.

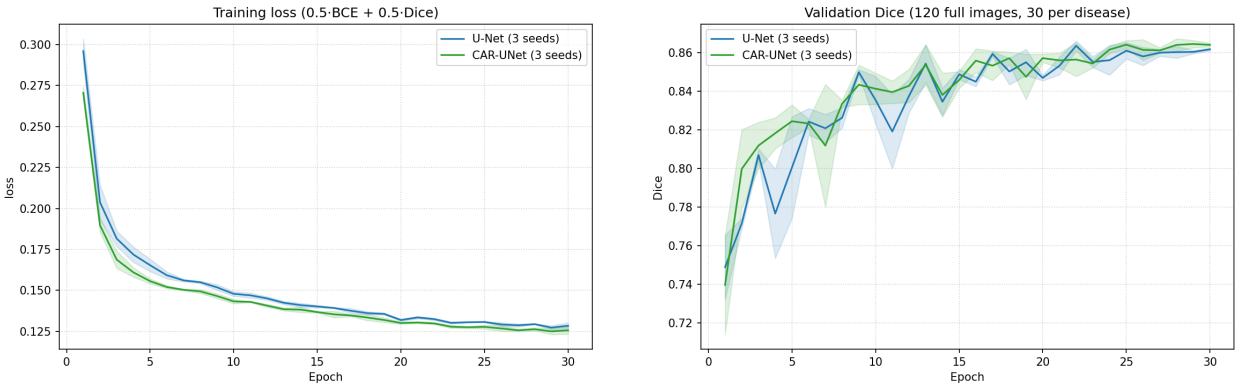


Figure 5: Training loss (left) and validation Dice (right) of U-Net and CAR-UNet, mean ± standard deviation over three seeds. Validation Dice levels off at about 0.86 over the last ten epochs.

Table 4
Structure of the code base (public repository, see the data and code availability statement).

Module	Purpose
`src/preprocessing_fives.py`	green channel, CLAHE, bilateral filter, FOV mask
`src/data_loader_fives.py`	disease-stratified split, patch sampling, augmentation
`src/car_unet.py`	CAR-UNet with MECA or dual attention
`src/losses.py`	BCE + Dice loss, soft-skeleton clDice loss
`src/metrics.py`	FOV-restricted metrics and clDice
`src/utils.py`	overlap-tile full-image inference
`scripts/train_fives.py`	training loop, model selection, test evaluation
`scripts/evaluate_post.py`	threshold selection and test-time augmentation
`scripts/summarize_*.py`	result tables and Wilcoxon tests

4.6. Evaluation

For every test image we count true positives (TP), false positives (FP), false negatives (FN) and true negatives (TN) inside the FOV, at a threshold of 0.5 unless stated otherwise, and compute

$$\begin{aligned} \mathrm{Dice} &= \frac{2TP}{2TP+FP+FN}, \qquad \mathrm{IoU} = \frac{TP}{TP+FP+FN}, \\ \mathrm{Sen} &= \frac{TP}{TP+FN}, \qquad \mathrm{Spe} = \frac{TN}{TN+FP}, \qquad \mathrm{Pre} = \frac{TP}{TP+FP}, \end{aligned} \tag{2}$$

together with accuracy, the area under the ROC curve (AUC) and the area under the precision–recall curve (AUC-PR). To measure connectivity we also compute the centreline Dice (Shit et al., 2021). With S_P and S_L the skeletons of the predicted and ground-truth masks V_P and V_L,

$$T_{\mathrm{prec}} = \frac{|S_P \cap V_L|}{|S_P|}, \qquad T_{\mathrm{sens}} = \frac{|S_L \cap V_P|}{|S_L|}, \qquad \mathrm{clDice} = \frac{2\,T_{\mathrm{prec}}\,T_{\mathrm{sens}}}{T_{\mathrm{prec}}+T_{\mathrm{sens}}}. \tag{3}$$

Because every vessel contributes by the length of its centreline rather than its area, clDice is sensitive to missing and broken thin vessels. Metrics are averaged over the 200 test images, and we report the mean and standard deviation over the three seeds. Differences between two models are tested with the paired Wilcoxon signed-rank test (Wilcoxon, 1945) on per-image scores averaged over seeds; this test is paired and does not assume normally distributed differences.

4.7. Proposed modifications

Our replication and the paper itself reveal five weaknesses: (W1) the attention only decides *which* feature channels matter, not *where* the vessels are; (W2) the pixel-wise loss hardly penalises a broken thin vessel; (W3) at 512×512 the

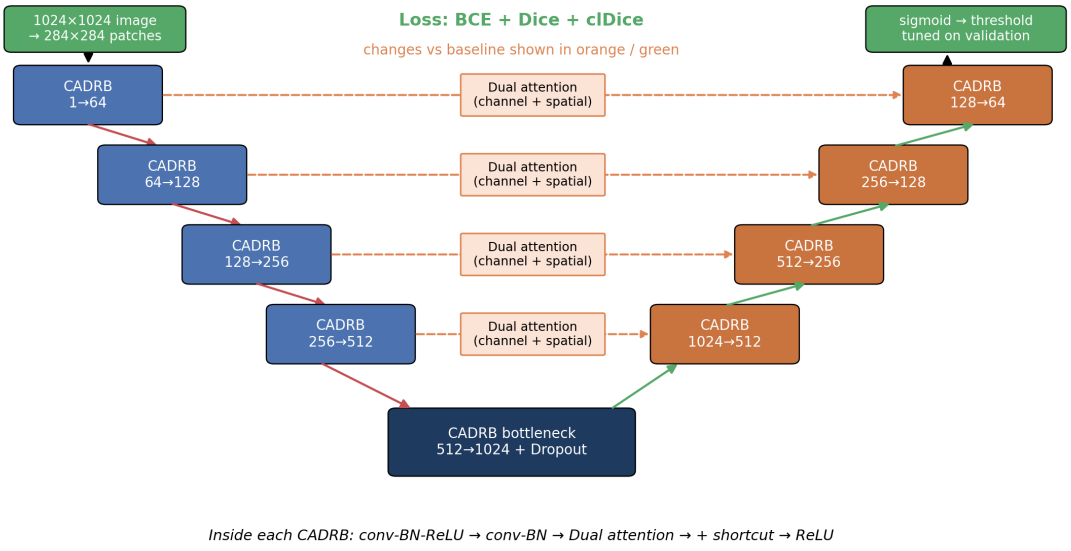


Figure 6: The modified architecture (changes in orange and green). Every attention module is replaced by dual attention, the input is 1024×1024, the loss includes clDice and the threshold is selected on the validation set.

thinnest capillaries are lost; (W4) a fixed decision threshold of 0.5 is used; and (W5) the evaluation is weak (few images, a single run, no overlap or connectivity measures). The protocol of Section 3 addresses W5. We address W1–W4 with the four modifications below, which are added cumulatively. Everything else, including data, split, seeds and training settings, is kept unchanged. Fig. 6 shows the modified architecture.

M1: Dual attention (W1). All 13 attention modules (nine inside the residual blocks and four on the skip connections) are replaced by a dual attention block. Its first step is channel attention computed from both the average- and the max-pooled channel descriptors with a shared 1D convolution, as in the paper's own MECA formulation. Max pooling keeps the strong but spatially small responses of thin vessels, which barely change a channel's average. Its second step is spatial attention in the style of CBAM (Woo et al., 2018), which assigns a weight to every pixel:

$$\mathbf{F}' = \sigma\big(\mathrm{C1D}_k(\mathrm{AvgPool}(\mathbf{F})) + \mathrm{C1D}_k(\mathrm{MaxPool}(\mathbf{F}))\big) \odot \mathbf{F}, \tag{4}$$

$$\mathbf{F}'' = \sigma\big(f^{7\times7}\big([\mathrm{Avg}_c(\mathbf{F}'); \mathrm{Max}_c(\mathbf{F}')]\big)\big) \odot \mathbf{F}', \tag{5}$$

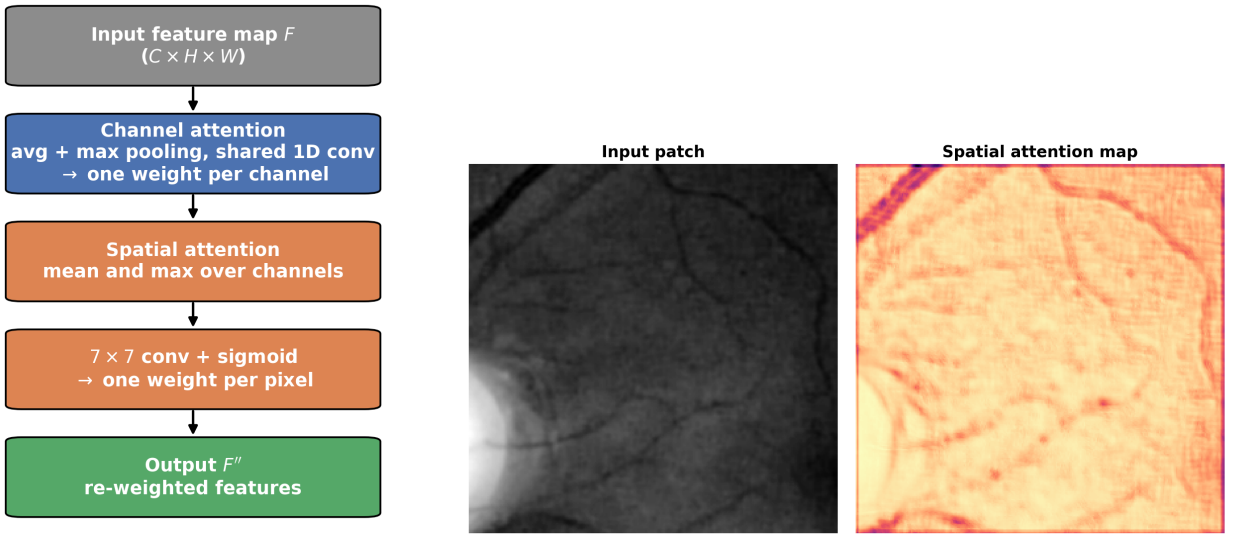
where Avg_c and Max_c are the mean and maximum over channels at each pixel and $f^{7\times7}$ is a 7×7 convolution (Fig. 7a). All 13 blocks together add only 1,274 parameters (+0.004%). Fig. 7b shows that the learned spatial attention map is organised around the vessel tree, including many thin branches; the block therefore re-weights the features according to vessel location, which channel attention alone cannot do.

M2: clDice loss (W2). We add the clDice loss (Shit et al., 2021), $\mathcal{L}_{\mathrm{clDice}} = 1 - \mathrm{clDice}$, computed with a differentiable soft skeleton (iterated soft erosion and dilation) on the predicted probabilities:

$$\mathcal{L} = 0.5\,\mathcal{L}_{\mathrm{BCE}} + 0.5\,\big[(1-\alpha)\,\mathcal{L}_{\mathrm{Dice}} + \alpha\,\mathcal{L}_{\mathrm{clDice}}\big], \qquad \alpha = 0.5. \tag{6}$$

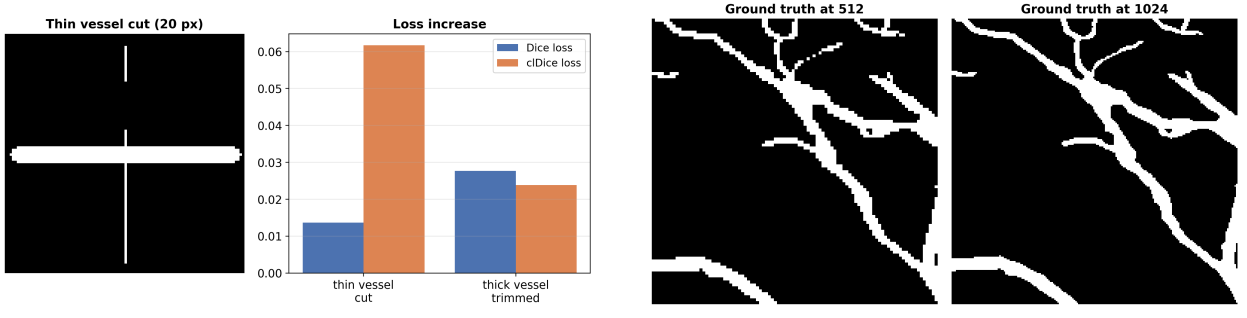
On a synthetic test image, cutting a thin vessel (20 pixels) raises the clDice loss 4.5 times more than the Dice loss, whereas trimming the end of a thick vessel costs less under clDice, because the vessel stays connected (Fig. 8a).

M3: Higher working resolution (W3). The improved model is trained and applied at 1024×1024 pixels. A capillary that is a fraction of a pixel wide at 512×512 becomes about twice as wide and survives the resizing (Fig. 8b). For a fair comparison, the 1024×1024 prediction is downsampled and scored on exactly the same 512×512 ground truth as the baseline. In addition, we score all models against the 1024×1024 ground truth, upsampling the predictions of the 512×512 models. The soft skeleton uses 20 instead of 10 iterations because vessels are twice as wide in pixels. The number of training patches per epoch is unchanged.



(a) (b)

Figure 7: Dual attention. (a) Structure of the block: channel attention from average and max pooling, followed by spatial attention that assigns one weight to every pixel. (b) A preprocessed input patch (glaucoma test image, 1024×1024 model) and the spatial attention map of the first encoder block (light = high weight). The vessel tree is visible in the map; in this early block, vessel pixels receive slightly lower weights than the background (mean 0.85 versus 0.90).



(a) (b)

Figure 8: Motivation for M2 and M3. (a) Synthetic test: cutting a 20-pixel piece out of a thin vessel raises the clDice loss 4.5 times more than the Dice loss, while trimming a thick vessel costs about the same under both. (b) The same ground-truth region at 512×512 and 1024×1024: thin vessels that are fragmented at 512×512 are continuous at 1024×1024.

M4: Validation-selected threshold (W4). For each trained model we evaluate thresholds from 0.20 to 0.80 (step 0.02) on the validation set and keep the one with the highest validation Dice. We also tested test-time augmentation (TTA), averaging the predictions on four flipped copies of each image, and let the validation set decide whether to use it.

5. Results

5.1. Comparison with the replicated baseline

Table 5 compares the replicated CAR-UNet with the improved model on the same split, with the same seeds and training settings. The improved model is better on Dice (+0.54 points), IoU (+0.83), sensitivity (+1.55), accuracy, AUC, AUC-PR and clDice (+1.25), and all of these differences are statistically significant. It is better in every one of the three seeds: Dice gains of +0.55, +0.78 and +0.30 points, and clDice gains of +0.68, +1.82 and +1.24 points.

Table 5
Replicated baseline versus improved model on the 200 FIVES test images (mean ± standard deviation over three seeds). Change in percentage points; *better on*: test images on which the improved model scores higher; p: paired Wilcoxon signed-rank test.

Metric	Baseline (CAR-UNet)	Improved	Change	Better on	p
Dice	0.8382 ± 0.0031	$\mathbf{0.8436 \pm 0.0027}$	$+0.54$	140/200	4.8×10^{-10}
IoU	0.7295 ± 0.0043	$\mathbf{0.7379 \pm 0.0039}$	$+0.83$	140/200	2.0×10^{-10}
Sensitivity	0.8310 ± 0.0126	$\mathbf{0.8465 \pm 0.0011}$	$+1.55$	159/200	7.4×10^{-19}
Specificity	$\mathbf{0.9855 \pm 0.0018}$	0.9846 ± 0.0008	-0.09	51/200	6.3×10^{-15}
Precision	$\mathbf{0.8557 \pm 0.0151}$	0.8516 ± 0.0054	-0.40	68/200	9.6×10^{-8}
Accuracy	0.9728 ± 0.0007	$\mathbf{0.9737 \pm 0.0006}$	$+0.09$	130/200	1.8×10^{-8}
AUC	0.9860 ± 0.0014	$\mathbf{0.9878 \pm 0.0003}$	$+0.17$	168/200	1.4×10^{-21}
AUC-PR	0.9202 ± 0.0044	$\mathbf{0.9260 \pm 0.0006}$	$+0.58$	163/200	9.4×10^{-18}
clDice	0.8626 ± 0.0057	$\mathbf{0.8751 \pm 0.0005}$	$+1.25$	175/200	5.3×10^{-25}

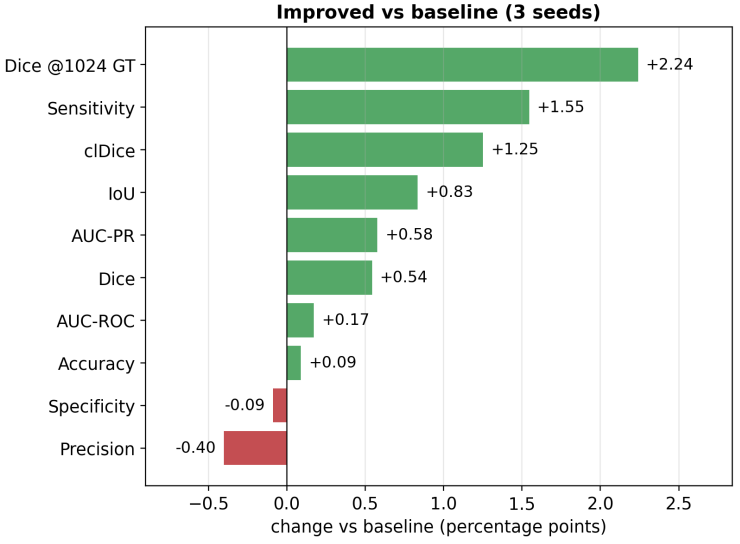


Figure 9: Change of every metric from the replicated baseline to the improved model (percentage points, three seeds). Green: the improved model is better.

Precision (-0.40) and specificity (-0.09) are slightly lower, which is the expected cost of detecting more vessel pixels. Fig. 9 summarises the change in every metric.

5.2. Ablation study

Table 6 and Fig. 10 add the modifications one at a time. Each component has a distinct effect. Dual attention raises Dice by 0.26 points, with a higher Dice on 167/200 images ($p = 2 \times 10^{-19}$), mainly by removing false detections (precision +0.63 points); it also gives the most stable results across seeds. The clDice loss improves connectivity (+0.88 clDice points over dual attention, better on 191/200 images, $p = 2 \times 10^{-32}$) and sensitivity (+1.24 points), but at the default threshold it also adds false detections, so Dice stays level. Moving to 1024×1024 raises Dice by 0.40 points over the same model at 512×512 (better on 150/200 images, $p = 2 \times 10^{-11}$) and precision by 2.0 points, at the cost of some sensitivity. Finally, the validation-selected threshold (0.44–0.48 for the three high-resolution models) converts this precision into recall: sensitivity rises by 1.39 points and Dice by 0.15 points (better on 148/200 images, $p = 5 \times 10^{-17}$).

Table 6
Ablation study: modifications added one at a time to the replicated CAR-UNet (200 test images, three seeds; Dice: mean ± standard deviation).

Model	Dice	IoU	Sensitivity	Precision	clDice
U-Net (reference)	0.8344 ± 0.0034	0.7246	0.8247	0.8573	0.8593
CAR-UNet (replicated baseline)	0.8382 ± 0.0031	0.7295	0.8310	0.8557	0.8626
+ M1 dual attention	0.8408 ± 0.0021	0.7339	0.8308	0.8620	0.8631
+ M2 clDice loss	0.8382 ± 0.0050	0.7293	0.8432	0.8430	0.8718
+ M3 1024×1024 input	0.8421 ± 0.0028	0.7359	0.8326	**0.8632**	0.8742
+ M4 tuned threshold (final)	$\mathbf{0.8436 \pm 0.0027}$	**0.7379**	**0.8465**	0.8516	**0.8751**

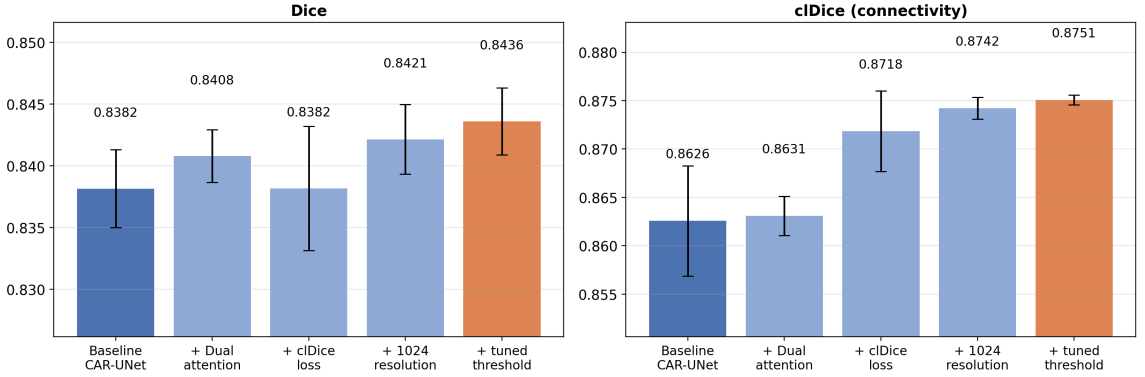


Figure 10: Ablation: Dice (left) and clDice (right) after each modification, mean ± standard deviation over three seeds.

Table 7
Dice and clDice per disease group (50 test images each, three seeds).

	Dice			clDice		
Disease	Baseline	Improved	Change	Baseline	Improved	Change
AMD	0.8643	**0.8671**	+0.28	0.8884	**0.8978**	+0.94
DR	0.8562	**0.8585**	+0.22	0.8807	**0.8895**	+0.88
Glaucoma	**0.7958**	0.7947	−0.11	0.8178	**0.8227**	+0.49
Normal	0.8363	**0.8541**	+1.78	0.8635	**0.8904**	+2.69

5.3. Per-disease results

Table 7 breaks the results down by disease group. Connectivity improves in every group. Dice improves for AMD, DR and normal eyes, most strongly for normal eyes (+1.78 points), but not for glaucoma (−0.11 points), which remains the hardest group for both models. Glaucoma images in FIVES contain the fewest annotated vessels (8.5% of the field of view, against 11.4% for normal eyes), so each missed vessel weighs more. The group also contains the two test images on which every model fails (Dice below 0.1): hazy, dark images whose expert masks contain only 0.7% and 1.7% vessel pixels, because few vessels are visible at all. We kept both images in the test set, since excluding hard cases would inflate the results.

5.4. Thin vessels

Two measurements focus on the thin vessels that motivated this work. First, the fraction of thin-vessel pixels (vessel radius ≤ 1.5 pixels in the ground truth) found by the model rises from 0.781 to 0.788 for seed 0, and is higher on 139 of 200 images ($p = 2.7 \times 10^{-7}$). Second, when all models are scored against the 1024×1024 ground truth, in which thin vessels are better represented, the 1024×1024 model reaches a Dice of 0.8796 ± 0.0015 and a clDice of 0.8972 (threshold 0.5), against 0.8572 and 0.8791 for the replicated CAR-UNet and 0.8599 and 0.8895 for the same model

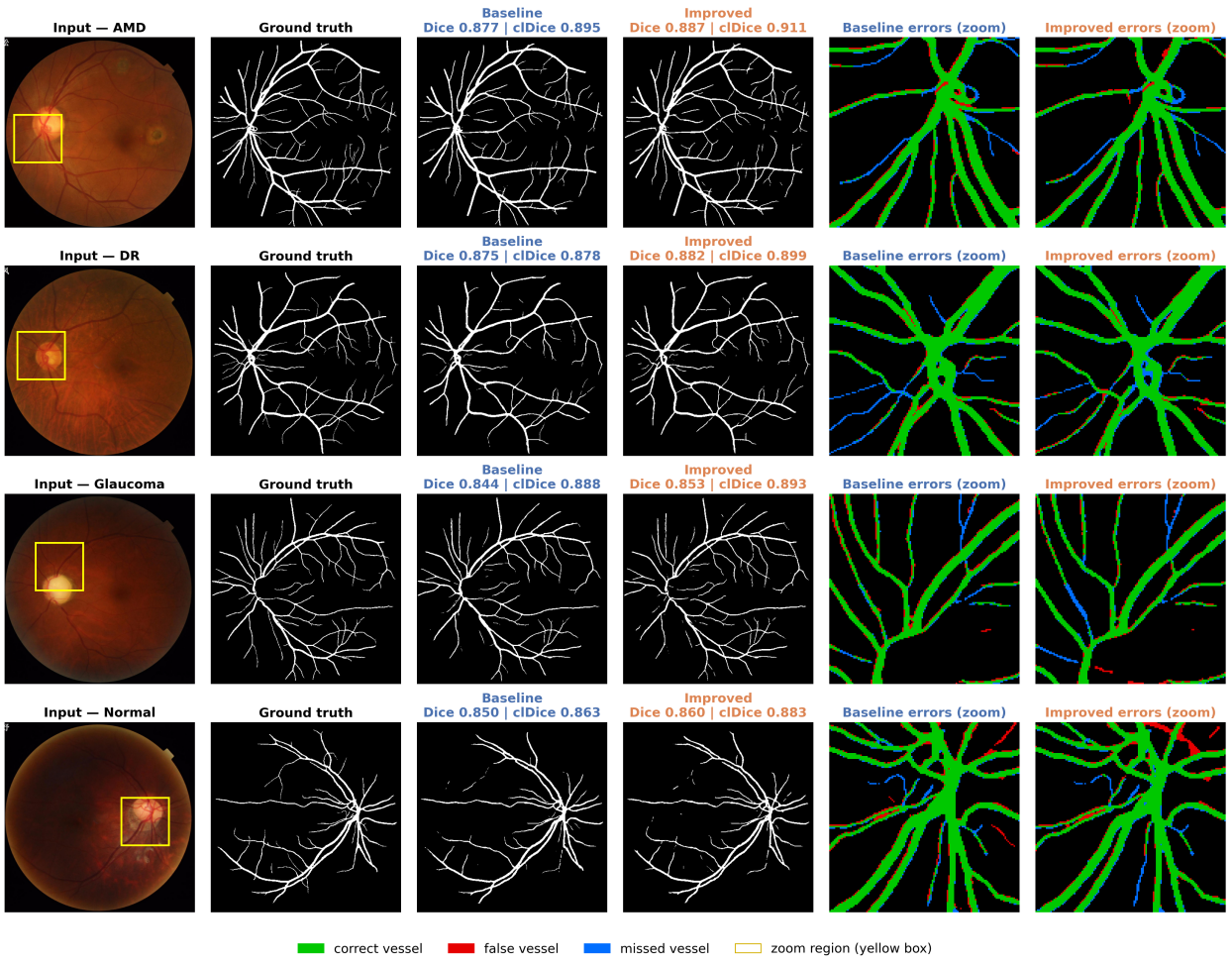


Figure 11: Qualitative comparison on one test image per disease group (the image with the median baseline Dice of its group, not hand-picked; seed 0). Columns: input (the yellow box marks the 128×128 region with the most thin vessels), ground truth, baseline and improved predictions, and zoomed error maps of both models (green: correct vessel, red: false vessel, blue: missed vessel).

trained at 512×512. Higher resolution therefore accounts for most of the gain against the 1024×1024 ground truth (+2.2 Dice points over the baseline).

5.5. Qualitative results

Fig. 11 shows the median test image of each disease group processed by both models. At full scale the differences are subtle, since both models trace the main vessels well, but the zoomed error maps reveal them: the improved model leaves fewer missed thin branches (blue) and keeps them connected, at the cost of a few additional false-positive pixels (red) along vessel borders. The improved model scores higher on all four examples, including the glaucoma image, although on average it does not improve Dice for that group (Table 7).

5.6. What did not work

Test-time augmentation lowered Dice for every model at the default threshold, by 0.2 to 0.7 points. A diagnostic on four validation images showed why: for one model, the individual predictions on flipped images were up to 2.6 Dice points worse than on the original image (0.868 original, 0.864 horizontal flip, 0.851 vertical flip, 0.842 both). Averaging them therefore mixes in weaker predictions. The models are not fully orientation-invariant, even though flips were used during training. Because the prediction setting is selected on the validation set, TTA was used for only

one of the three final models. Threshold tuning, in turn, helped only the 1024×1024 models; for the 512×512 models it changed Dice by less than 0.1 points, so they were already well calibrated at 0.5.

6. Discussion

Our findings demonstrate that the remaining errors of CAR-UNet on multi-disease data lie mainly in thin vessels and connectivity, and that changes aimed at them improve the model consistently. Each modification acts on a different part of the problem, and the ablation shows that their effects are complementary: spatial attention lets the network weight its features by location, the clDice loss makes broken vessels expensive during training, the higher resolution keeps capillaries visible in the first place, and the tuned threshold balances precision and recall at the end. The gains are larger on the measures that target thin vessels (sensitivity, clDice, Dice against the 1024×1024 ground truth) than on Dice at 512×512, which is dominated by thick vessels that both models already segment well.

These observations agree with earlier work. Combining channel and spatial attention helped in natural images (Woo et al., 2018), and the authors of CAR-UNet had already used spatial attention alone in SA-UNet (Guo et al., 2021b); our results suggest that the two forms of attention are complementary for vessels. The clDice loss improved connectivity much more than overlap, which matches the purpose for which it was designed (Shit et al., 2021). Absolute values cannot be compared with those of the base paper, because the datasets differ, but our replicated accuracy, specificity and AUC fall in the range the paper reports.

One could argue that the gain comes only from the higher resolution. The ablation does not support this: at 512×512, dual attention alone raised Dice on 167 of 200 images, and the clDice loss raised connectivity on 191 of 200 images. A second objection is that a gain of 0.54 Dice points is small. It is, but it is consistent: the improved model is better in all three seeds and on 140 of 200 test images, and its seed-to-seed standard deviation is lower than that of the baseline. Single-seed differences between models in our replication varied by up to one Dice point, which is why multiple seeds and paired tests were necessary to separate real effects from training noise.

Limitations. (i) Our CAR-UNet differs from the original in its convolution geometry, width and regulariser, so absolute numbers are not directly comparable with the paper. (ii) All experiments use a single dataset; we did not test generalisation to images from other cameras. (iii) Glaucoma images are not improved, and two very low-quality images fail for every model. (iv) Because of the higher resolution, the improved model needs about twice the training time of the baseline (about 195 versus 91 minutes per run) and four times as many tiles at inference. (v) The slightly lower precision may matter in applications where false vessels are costly.

The take-home message is that, for vessel segmentation, the model's attention, its loss and its input resolution should be chosen with the thinnest vessels in mind, and that such improvements should be judged over several seeds with connectivity measures, not by pixel accuracy alone.

7. Conclusion

We replicated CAR-UNet on the multi-disease FIVES dataset under a strict, three-seed protocol and confirmed the paper's claim that it outperforms U-Net, albeit by a small margin. Motivated by an analysis of its errors, we added dual attention, a clDice loss, a higher working resolution and a validation-selected threshold. The improved model reaches a Dice score of 0.8436, an IoU of 0.7379 and a clDice of 0.8751, outperforms the replicated baseline on nine of eleven metrics with statistically significant differences, and gains most where it matters clinically: thin vessels and vessel connectivity. Future work should focus on the glaucoma group and on low-quality images, for example by using the image-quality labels provided with FIVES, explore training at the full 2048×2048 resolution, and investigate lighter networks closer to the original 16-filter design for faster screening.

CRediT authorship contribution statement

Abhishek Karthik J: Conceptualization, Methodology, Software, Investigation, Formal analysis, Visualization, Writing – original draft, Writing – review and editing. **Abhishek Ramesh:** Conceptualization, Methodology, Software, Investigation, Formal analysis, Visualization, Writing – original draft, Writing – review and editing. **Krish S:** Conceptualization, Methodology, Software, Investigation, Formal analysis, Visualization, Writing – original draft, Writing – review and editing. **Theerth Krish S:** Conceptualization, Methodology, Software, Investigation, Formal analysis, Visualization, Writing – original draft, Writing – review and editing.

All four authors contributed equally to this work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial or not-for-profit sectors.

Ethics statement

This study uses only the publicly released, de-identified FIVES dataset (Jin et al., 2022). No new data were collected from human participants, so no additional ethics approval was required.

Data and code availability

FIVES is publicly available (Jin et al., 2022). The code, the per-image evaluation results and all scripts needed to reproduce the reported numbers are available at
https://github.com/Abhishek-karthik/Retinal-Vessel-Segmentation-CAR-UNet.

References

Frangi, A.F., Niessen, W.J., Vincken, K.L., Viergever, M.A., 1998. Multiscale vessel enhancement filtering, in: Medical Image Computing and Computer-Assisted Intervention (MICCAI), pp. 130–137.

Ghiasi, G., Lin, T.Y., Le, Q.V., 2018. DropBlock: A regularization method for convolutional networks, in: Advances in Neural Information Processing Systems (NeurIPS).

Guo, C., Szemenyei, M., Hu, Y., Wang, W., Zhou, W., Yi, Y., 2021a. Channel attention residual U-Net for retinal vessel segmentation, in: IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 1185–1189. doi:10.1109/ICASSP39728.2021.9414282.

Guo, C., Szemenyei, M., Yi, Y., Wang, W., Chen, B., Fan, C., 2021b. SA-UNet: Spatial attention U-Net for retinal vessel segmentation, in: 25th International Conference on Pattern Recognition (ICPR), pp. 1236–1242. doi:10.1109/ICPR48806.2021.9413346.

He, K., Zhang, X., Ren, S., Sun, J., 2016. Deep residual learning for image recognition, in: IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 770–778.

Hu, J., Shen, L., Sun, G., 2018. Squeeze-and-excitation networks, in: IEEE Conference on Computer Vision and Pattern Recognition (CVPR), pp. 7132–7141.

Jin, K., Huang, X., Zhou, J., Li, Y., Yan, Y., Sun, Y., Zhang, Q., Wang, Y., Ye, J., 2022. FIVES: A fundus image dataset for artificial intelligence based vessel segmentation. Scientific Data 9, 475. doi:10.1038/s41597-022-01564-3.

Kingma, D.P., Ba, J., 2015. Adam: A method for stochastic optimization, in: International Conference on Learning Representations (ICLR).

Liskowski, P., Krawiec, K., 2016. Segmenting retinal blood vessels with deep neural networks. IEEE Transactions on Medical Imaging 35, 2369–2380.

Loshchilov, I., Hutter, F., 2017. SGDR: Stochastic gradient descent with warm restarts, in: International Conference on Learning Representations (ICLR).

Milletari, F., Navab, N., Ahmadi, S.A., 2016. V-Net: Fully convolutional neural networks for volumetric medical image segmentation, in: Fourth International Conference on 3D Vision (3DV), pp. 565–571.

Oktay, O., et al., 2018. Attention U-Net: Learning where to look for the pancreas. ArXiv:1804.03999.

Ronneberger, O., Fischer, P., Brox, T., 2015. U-Net: Convolutional networks for biomedical image segmentation, in: Medical Image Computing and Computer-Assisted Intervention (MICCAI), pp. 234–241.

Shit, S., Paetzold, J.C., Sekuboyina, A., Ezhov, I., Unger, A., Zhylka, A., Pluim, J.P.W., Bauer, U., Menze, B.H., 2021. clDice – a novel topology-preserving loss function for tubular structure segmentation, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 16555–16564.

Staal, J., Abràmoff, M.D., Niemeijer, M., Viergever, M.A., van Ginneken, B., 2004. Ridge-based vessel segmentation in color images of the retina. IEEE Transactions on Medical Imaging 23, 501–509.

Tomasi, C., Manduchi, R., 1998. Bilateral filtering for gray and color images, in: Sixth International Conference on Computer Vision (ICCV), pp. 839–846.

Wang, Q., Wu, B., Zhu, P., Li, P., Zuo, W., Hu, Q., 2020. ECA-Net: Efficient channel attention for deep convolutional neural networks, in: IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 11531–11539.

Wilcoxon, F., 1945. Individual comparisons by ranking methods. Biometrics Bulletin 1, 80–83.

Woo, S., Park, J., Lee, J.Y., Kweon, I.S., 2018. CBAM: Convolutional block attention module, in: European Conference on Computer Vision (ECCV), pp. 3–19.

Zuiderveld, K., 1994. Contrast limited adaptive histogram equalization, in: Graphics Gems IV. Academic Press, pp. 474–485.